Geotectonics of Cretaceous and Eocene plutons in British Columbia: a paleomagnetic fold test

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A total of 295 cores (590 specimens) were collected at 59 sites in the Coast plutonic complex along an E–W section southwest of Kitimat, British Columbia. The sites represent the Ponder, Alastair Lake, and Quottoon plutons in the 40–50 Ma eastern K–Ar age zone and the Ecstall and Butedale plutons in the 64–80 Ma central age zone. After af demagnetization a stable remanent magnetization was isolated at 32 sites and these data were combined with available data from the Skeena River section about 100 km to the north. The remanence directions in sites from the NNW-trending north and south limbs of the Hawkesbury Warp provide a positive fold test when compared to the WNW-trending centre limb directions.

In the Eocene eastern age zone the NNW limbs give a concordant pole position relative to the cratonic North American pole whereas the centre limb has undergone ≈50° of the counter-clockwise rotation and ≈10° of upward tilt of its western end to give a discordant pole. In the late Upper Cretaceous central age zone, the Ecstall–Butedale pluton was tilted 15° to the west on all limbs before the Eocene intrusion and Hawkesbury Warp deformation events to give a NNW-trend pole and WNW-trend pole diverging in opposite directions from the cratonic reference pole.

The geologic field evidence from structural trends, from fault, fold, contact, and foliation attitudes, and from distribution of plutonic phases is consistent with the structural model. The regional geotectonic events are related to possible Cenozoic plate interactions on the western margin of the North American plate. This combination of concordant and discordant poles cannot be explained in terms of an excursion of the geomagnetic paleopole during intrusion, a large scale northward translation of the western Cordillera during the Cenozoic, or a combination of clockwise rotations and northward translations on the margin of the advancing North American plate. The fold test and polarity reversal pattern indicate that all plutons acquired a primary thermoremanent magnetization (TRM) during cooling and probably within ≈1 Ma after emplacement.

On a prélevé un total de 295 carottes (590 spécimens) à 59 sites dans le complexe plutonique Côtier le long d'une section E–O au sud-ouest de Kitimat, Colombie-Britannique. Les sites représentent les plutons de Ponder, d'Alastair Lake et de Quottoon dans la zone chronologique est datant de 40 à 50 Ma (K–Ar) et les plutons d'Ecstall et de Butedale dans la zone d'âge centrale de 64 à 80 Ma. Après démagnétisation af, on a isolé une magnétisation rémanente stable à 32 sites et on a combiné ces résultats avec les données disponibles de la section de Skeena River, environ 100 km au nord. Les directions de rémanence des sites localisés dans les flancs nord et sud de la voûte de Hawkesbury orientés NNO permettent d'obtenir un indice positif de plissement si on les compare aux directions de la partie centrale orientée ONO.

Dans la zone d'âge à l'est datant de l'Eocène, les flancs orientés NNO donnent une position du pôle qui concorde avec le pôle cratonique nord-américain alors que la partie centrale a subi une rotation négative de 50° environ et un basculement de 10° environ vers le haut dans sa partie ouest pour donner un pôle discordant. Dans la zone centrale d'âge, datant de la fin du Crétacé supérieur, le pluton Ecstall–Butedale a basculé de 15° vers l'ouest sur tous ses flancs avant l'intrusion de l'Eocène et la déformation de la voûte de Hawkesbury pour donner un pôle de direction ONO et un pôle de direction NNO qui divergent dans des directions opposées du pôle de référence du craton.

Les observations géologiques sur le terrain concernant les orientations structurales, les failles, les plis, les contacts, les attitudes de la foliation et la distribution des phases plutoniques sont en accord avec le modèle structural. Les événements géotectoniques régionaux s'expliquent par les interactions possibles des plaques durant le Cénozoïque le long de la bordure ouest de la plaque de l'Amérique du Nord. Cette combinaison de pôles concordants et discordants ne peut s'expliquer en termes du déplacement du paléopôle géomagnétique au cours de l'intrusion, d'une translation à grande échelle vers le nord de la Cordillère occidentale durant le Cénozoïque, ou d'une combinaison de rotation positives et de translations vers le nord à la bordure de la plaque de l'Amérique du Nord aussi en mouvement. L'indice de plissement et le comportement des renversements de polarité indiquent que tous les plutons ont acquis une TRM primaire durant leur refroidissement et probablement en dedans de ≈1 MA après leur mise en place.

[Traduit par le journal]

Introduction

The age of emplacement of the plutonic rocks that form the Coast plutonic complex remains a major uncertainty in the interpretation of Cordilleran geology. Geologic field relations place their time of intrustion between Late Jurassic and Late Miocene. Hutchison (1967, 1970) and Monger and Hutchison (1971) noted that the K–Ar radiometric ages from plutons along the length of the complex form an eastern zone of 40–50 Ma, a central zone of 64–80 Ma, and a western zone of 84–140 Ma. These may date: (*a*) the times of sequential plutonic emplacement; (*b*) the passage of a thermal metamorphic front from west to east; or (*c*) the cessation of argon diffusion upon unroofing during progressive uplift from west to east. In a recent paleomagnetic study of sites from Cenozoic plutons across the eastern and central age zones along the Skeena River, Symons (1974*a*) showed that the remanence directions are related to plutonic emplacement. The site directions do not record an alternating normal and reverse polarity profile which would result from an easterly-moving metamorphic front or from a progressive uplift.

This paleomagnetic study reports on the results from a further 59 sites along a 125-km section southwest from Kitimat, B.C. (Fig. 1). This section is about 100 km southeast of the Skeena River section. It also crosses the eastern and central age zones. Because many of the Skeena River section plutons have been further sampled, the results from that study are integrated into this discussion. The Kitimat section provides a unique location to apply a fold test (Graham 1949) of regional scale because the Coast plutonic trend has the form of an S.

Geology

The Kitimat section is located in the Douglas Channel – Hecate Strait map area described geologically by Roddick (1970) except for sites 1, 2, and 3 (Fig. 2) which are in the Terrace map area described by Duffell and Souther (1964).

The oldest rock units in the Kitimat section are metamorphosed sediments and volcanics ranging from greenschists on the east to amphibolites on the west. These strata are thought to correlate with unmetamorphosed Carboniferous, Triassic, and Jurassic strata outside the area. The youngest strata cut by the plutons are unmetamorphosed sediments of the Bowser group which contain late Jurassic or earliest Cretaceous fossils (Eisbacher 1973). Also granitic clasts from the plutons are absent in the Bowser group whereas they are present in the overlying Lower Cretaceous sediments.

Hutchison (1970) has summarized the petrology and structure of the central Coast plutonic complex. Individual plutons are elongate, often tadpole-shaped, and they parallel the NNW trend of the complex. They may exceed 100 km in length as does the Quottoon pluton, or 30 km in width as do the Ponder and Butedale plutons. The plutons generally have a perimeter phase of foliated or gneissic diorite with minor gabbroic zones. Progressing towards their core they become increasingly massive and grade from quartz diorite, to granodiorite, to a massive quartz monzonite core.

The Ponder, Alastair Lake, and Quottoon plutons lie in the eastern age zone. The Ponder pluton gives K–Ar ages on biotite of 46 $\pm$10 and 47 $\pm$4 Ma (GSC 65-32 and 66-15 in Wanless *et al.* (1967, 1968)). The Alastair Lake pluton gives a biotite K–Ar age of 44 $\pm$4 Ma (GSC 65-30). It appears to be the southward equivalent of the Kasiks pluton which was previously sampled (Symons 1974*a*) and gives a biotite K–Ar age of 43 $\pm$5 Ma (GSC 65-29). The Quottoon pluton gives biotite and hornblende K–Ar ages of 44 $\pm$5, 48 $\pm$9, 49 $\pm$4, 49 $\pm$7, and 50 $\pm$5 Ma (GSC 66-6, 7, 8, and 9; and GSC 67-25 in Wanless *et al.* (1970)); its metamorphic halo gives an age 48 $\pm$5 Ma (GSC 64-12 in Wanless *et al.* (1966)).

The Ecstall and Butedale plutons are in the central age zone. The Ecstall pluton narrows at its southernmost end and then expands to form the Butedale pluton. The Ecstall pluton gives biotite and hornblende K–Ar ages of 64 $\pm$8, 70 $\pm$4, and 74 $\pm$15 Ma (GSC 65-31, and 66-12 and 13) and the Butedale pluton gives a biotite K–Ar age of 67 $\pm$5 Ma (GSC 64-11).

Fresh undeformed diabase dikes cut plutons of the Coast complex (Hutchison 1967; Baer 1967). The dikes give late Miocene K–Ar ages of 12 $\pm$2, 12.5 $\pm$2.7, 13 $\pm$2, and 14.5 $\pm$1.0 Ma (GSC 67-31 and 32; and 70-7 and 8 in Wanless *et al.* (1970, 1972)) and a concordant Miocene paleomagnetic pole position (Symons 1969*a*). Roddick (1970) considers these dikes to be related to late Tertiary volcanic centres in the Douglas Channel – Hecate Strait map-area.

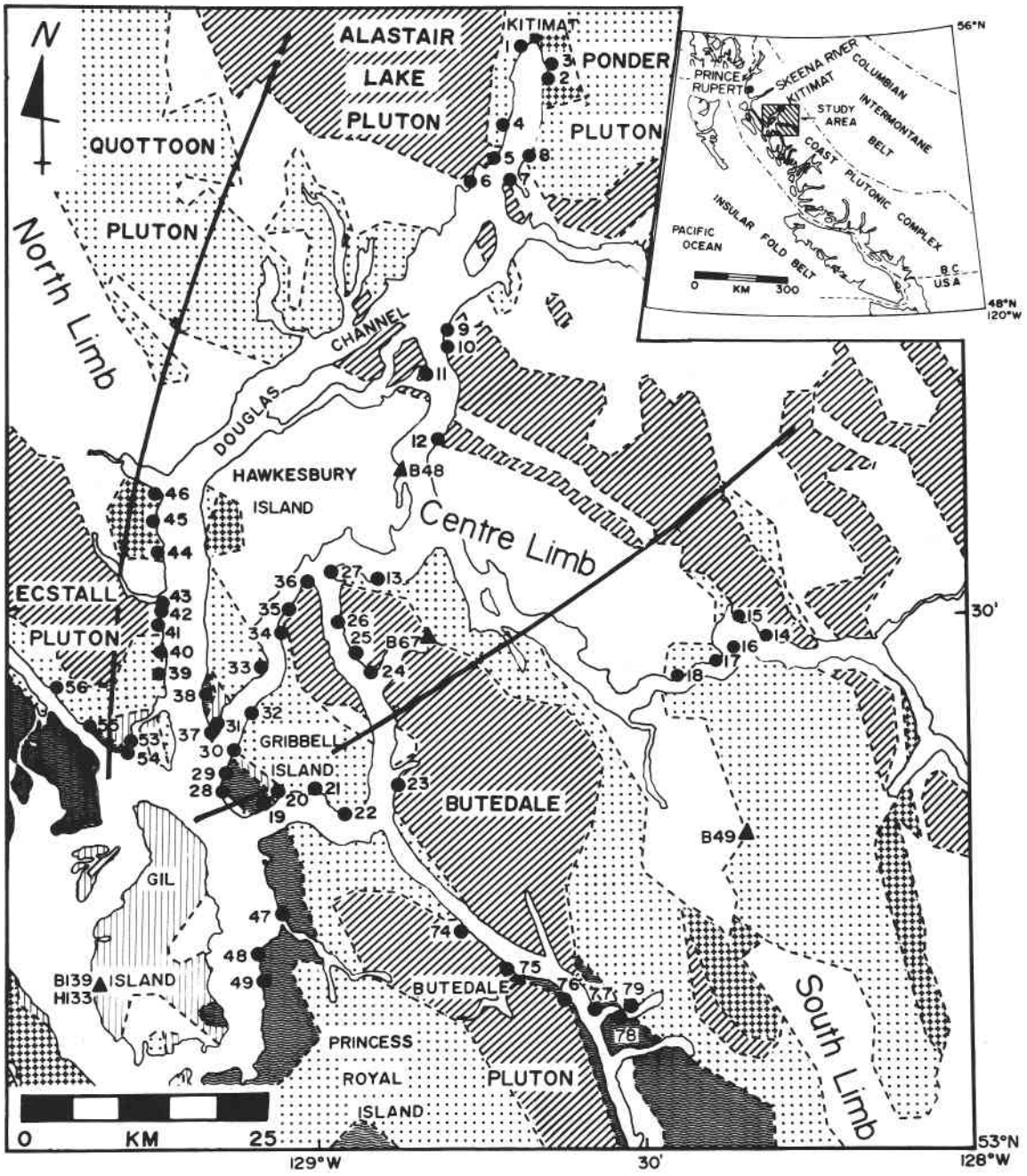


FIG. 1. Geology simplified from Roddick (1970) and site locations. Pre-Cretaceous metamorphic rocks are unpatterned. The plutonic phases are: gneissic diorite, horizontal wavy lines; hornblende diorite, vertical lines; quartz diorite, stipple; granodiorite, diagonal lines; and quartz monzonite, crosses. Sites are shown as solid circles with the number. Radiometric K–Ar age locations are shown as triangles with the method (B—biotite; H—hornblende) and age (Ma). The hinge lines of the Hawkesbury Warp are shown as thick solid lines.

TABLE 1. Remanence data of accepted sites

Site	Location °W	Location °N	Pluton and phase	AF intens. Oe	Data screening a	b	c	N	Remanent magnetization Length R	Decl.	Incl. +down	α_{95}
1	128.68	54.05	P– 8b	200				5	4.460	74.9	65.3	30.0
4	128.71	53.96	P– 8b	700	1			4	3.818	308.8	55.5	23.0
6	128.77	53.90	A– 9b	200				5	4.794	315.1	56.5	18.0
9	128.80	53.77	Q– 9c	200				5	4.843	268.9	47.2	15.0
10	128.81	53.76	Q– 9c	200				5	4.977	279.9	39.6	5.8
11	128.84	53.72	Q– 9c	200				5	4.521	119.8	−31.7	28.0
12	128.82	53.66	Q– 9c	150	1			4	3.573	311.0	29.5	37.0
13	128.91	53.53	B– 8b	200	2			3	2.991	108.8	−40.2	8.3
16	128.35	53.46	Q– 8b	700				5	4.931	73.4	−66.1	10.0
17	128.37	53.45	Q– 8a	200				51	4.178	192.0	−73.2	39.0
18	128.43	53.44	Q– 8a	200				5	4.755	190.5	−57.6	20.0
20	129.06	53.33	B– 7b	550				5	4.458	355.6	55.4	7.7
23	128.88	53.35	B– 9c	200	1			4	3.525	19.5	66.0	40.0
26	128.97	53.48	B– 9c	200				5	4.872	109.5	−48.1	14.0
28	129.15	53.33	B– 8b	200				5	4.980	352.0	57.9	5.4
29	129.14	53.34	B– 5b	700				5	4.844	353.2	61.7	14.0
30	129.13	53.36	B– 7b	550			1	4	3.520	7.1	66.0	40.0
31	129.12	53.38	B– 8b	700	1			4	3.936	341.2	53.2	14.0
32	129.10	53.40	B– 8b	700				5	4.881	158.1	−51.3	13.0
33	129.08	53.45	B– 8b	400				5	4.989	337.0	61.8	4.1
37	129.17	53.38	B– 7b	700			1	4	3.987	357.3	56.8	6.1
38	129.18	53.40	B– 5b	200				5	4.672	353.7	63.8	23.0
42	129.23	53.50	E– 9c	300	1			4	3.816	321.9	60.4	16.0
46	129.25	53.63	E– 10a	300	1			4	3.964	187.7	−55.7	10.0
47	129.05	53.22	B– 5b	200	1			4	3.838	29.4	61.1	22.0
53	129.28	53.37	E– 7b	700	1			4	3.881	328.6	47.4	19.0
54	129.29	53.36	E– 5b	300				5	4.944	336.6	61.0	9.1
55	129.35	53.35	E– 8b	100				5	4.893	18.4	76.7	13.0
56	129.39	53.42	E– 8b	300				5	4.939	55.4	61.7	9.5
74	128.77	53.21	B– 9c	200	2			3	2.890	0.6	63.9	30.0
75	128.71	53.17	B– 9c	300	1			4	3.944	1.8	50.6	13.0
78	128.53	53.12	B– 8b	200	1			4	3.879	29.3	80.4	19.0

NOTES: Location is given by longitude (°W) and latitude (°N). The pluton and phase abbreviations are: A, Alastair Lake; B, Butedale; E, Ecstall; P, Ponder; Q, Quottoon; 5b, gneissic diorite–migmatite complex; 7b, hornblende diorite; 8a, biotite–quartz diorite; 8b, hornblende–quartz diorite; 9b, biotite granodiorite; 9c, hornblende granodiorite; 10a, biotite–quartz monzonite. The intensity of the af demagnetizing field is given in oersteds. The data screening is described in the text. *R* is the length of the vector resultant. The declination, inclination, and radius of 95% confidence (Fisher 1953) are in degrees.

Experimental Methods

Sites 1, 2, and 3 are from road cuts; the remaining 56 sampling sites were collected by boat from shoreline exposures along some 300 km of fiord channels. The sites were distributed as uniformly as possible from the available outcrops of as many intrusive phases as possible (Fig. 1, Table 1).

Five well-separated cores were drilled at each site. The cores were oriented *in situ* to within 2° of arc using a sun compass with Brunton compass readings and topographic sightings as checks. Two specimens of 2.50-cm diameter and 2.40-cm length were cut from each core. The natural remanent magnetization (NRM) of every specimen was measured using a Schonstedt SSM-1A spinner magnetometer. One pilot specimen per site with an average NRM direction and intensity for that site was step-demagnetized in alternating fields (af) with intensities of 50, 100, 150, 200, 300, 400, 550, 700, 850, and 1000 Oe (4×10^3, 8×10^3, 12×10^3, 16×10^3, 24×10^3, 32×10^3, 44×10^3, 56×10^3, 68×10^3, and 80×10^3 A/m)[1] peak using a Schonstedt GSD-1 af demagnetizer. The paleomagnetic stability index (PSI) method of Symons and Stupavsky (1974) for directional analysis and the intensity decay curves for

[1] 1 Oe = 79.6 A/m in SI units.

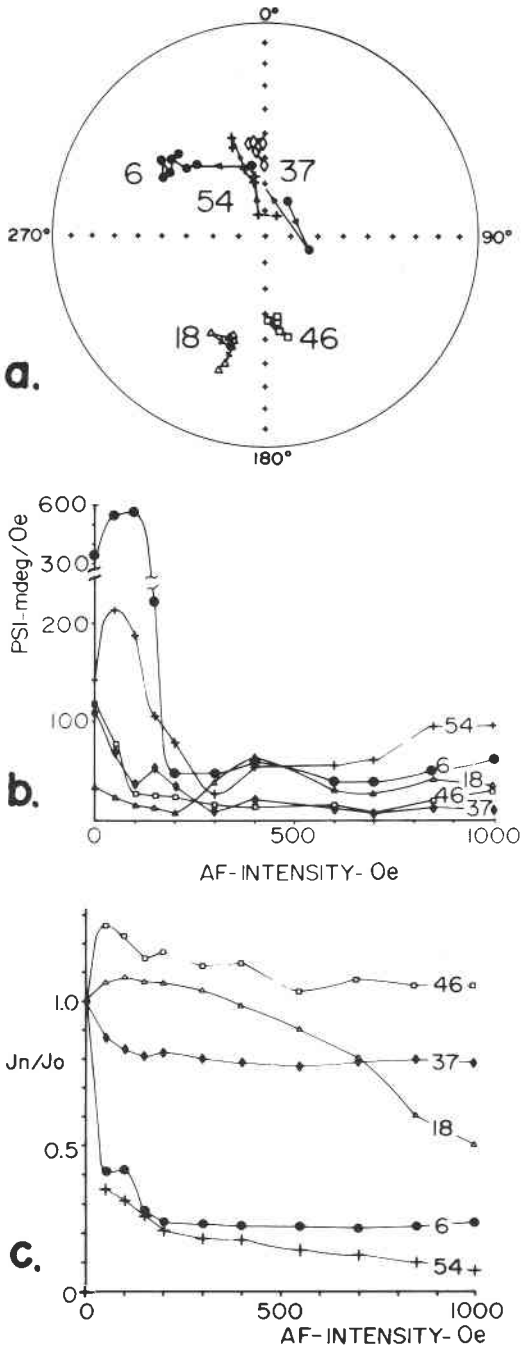


FIG. 2(*a*). Equal area projection showing the change in direction on progressive af demagnetization in fields of 0, 50, 100, 150, 200, 300, 400, 550, 700, 850, and 1000 Oe for a gneissic diorite (cross, 54), diorite (diamond, 37), quartz diorite (triangle, 18), granodiorite (circle, 6), and monzonite (square, 46) specimens. Solid (open) symbols indicate normal (reverse) or down (up) directions. (*b*) Paleomagnetic Stability Index (PSI) is in millidegrees/oersted and the af demagnetizing field (H) in oersteds. (*c*) Remanence intensity (J) is plotted as a ratio of the NRM intensity (J_0) against the af demagnetizing field in oersteds.

each pilot specimen were used to select optimum af intensity for demagnetizing the remaining specimens for each site (Table 1).

The pilot specimens exhibit a common behaviour on af step demagnetization. Figure 2*a* shows the directional changes for five typical specimens representing the different petrologic phases. The normally-magnetized specimens (6, 37, and 54) show clearly the removal of unstable viscous remanence (VRM) components in the present steeply-inclined earth's field direction in the 50 and 100 Oe (4×10^3 and 8×10^3 A/m) steps. Reversely-magnetized specimens (18 and 46) show the effect less clearly. However, when the PSI curves are examined (Fig. 2*b*), the effect is clear as all five specimens show high initial rates of directional change usually exceeding 100 mdeg/Oe. The PSI curves decrease noticeably to less than 20 mdeg/Oe between 200 and 700 Oe (16×10^3 and 56×10^3 A/m) as the stable direction is isolated. Thereafter the curves record an increasing rate of directional change in most specimens as larger random anhysteretic remanence (ARM) components are added in the demagnetizing process. The intensity decay curves support this interpretation (Fig. 2*c*). The normal specimens (6, 37, and 54) show an initial rapid drop in intensity up to 200 Oe (16×10^3 A/m) as normal VRM components are removed leaving the stable components. Conversely, the reverse specimens (18 and 46) show the expected initial intensity increase as the VRM components are subtracted from the stable reverse components. Above 200 Oe (16×10^3 A/m), most specimens show a relatively slow decrease in remanence intensity as the stable direction is isolated. Some specimens show slight increases near 1000 Oe (80×10^3 A/m) as ARM components are introduced into the specimens.

Specimens from 8 representative sites have been thermally demagnetized at 9 temperature steps up to 650°C (Fig. 3) using a nonmagnetic induction furnace in nested mu-metal shields in a shielded room. One specimen (site 4) proved to be either weakly or unstably magnetized above 400°C. When present, initial viscous components are removed by 150 or 300°C. All

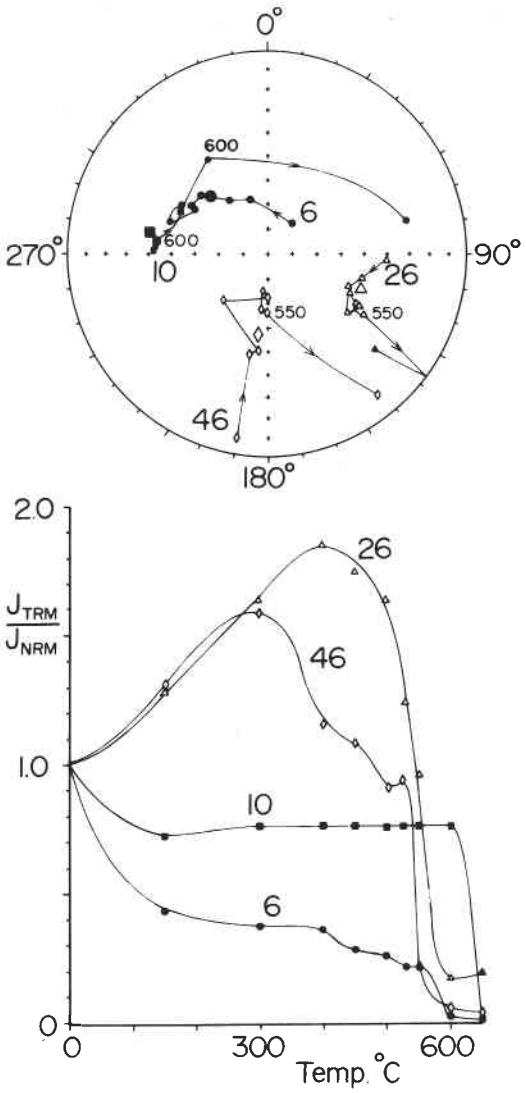


FIG. 3. Thermal demagnetization data for sites 6 (circles), 10 (squares), 26 (triangles), and 46 (diamonds). On the stereonet arrows indicate the direction of increasing temperature with the actual temperature in °C at some points. The larger symbol indicates the site mean direction from demagnetization. The graph shows the decay in the step remanence to initial natural remanence with temperature.

seven specimens show isolation of a single component to temperatures above 525°C (site 18), 550°C (sites 6 and 26), 600°C (sites 37, 46 and 54), and 650°C (site 10). The thermal intensity decay curves are knee-shaped showing that the thermal coercivity spectrum of the stable direction is blocked dominantly above 525°C. Following Pullaiah *et al.* (1975) the remanence is retained primarily in magnetite and to some extent in hematite and it has been stable through geologic times. It would have required a thermal metamorphic event in excess of mid-greenschist metamorphic levels to reset the primary magnetization. There is no evidence in the plutons for such metamorphism.

Statistical Analysis

The following tests were used to select reliable directions after af cleaning (Table 1). In 89 of 295 cores the directions for the two specimens diverged by more than 20°. These cores were considered to be inhomogenously magnetized and their data rejected.

Three sites then had only one acceptable core and so they were rejected as being insufficiently represented. The specimen directions were then combined to give the core mean directions which were then averaged to give the site mean directions. In 2 sites, the direction of one core deviated by $>40°$ from the site mean direction. Data from these two anomalous cores were rejected. Site mean directions with a radius of 95% confidence (α_{95}, Fisher 1953) exceeding 40° were regarded as insufficiently homogenously magnetized and rejected.

A total of 32 of the original 59 sites (54%) survived the screening tests and yielded accepted site mean directions. This is lower than the 64% acceptibility rate of the original Skeena River section study (Symons 1974*a*), probably because of the larger percentage of sites in more acidic plutonic phases. The survival rates are as follows: gneissic diorite (45%), hornblende diorite (100%), quartz diorite (63%), granodiorite (50%), and monzonite (20%).

In the Skeena River section (Symons 1974*a*), each plutonic phase has a well-defined cluster of site mean directions. With one exception, the radii for their cones of 95% confidence (α_{95}) are less than 11°. Plutonic phase mean directions were initially calculated (Fig. 4; Table 2) by averaging the site directions from both the Kitimat and Skeena River sections. The results did not conform to the previous Skeena River pattern and appeared to reflect the Hawkesbury Warp folding.

The Hawkesbury Warp has three limbs. The north limb trends NNW (N 25° W) and includes the Skeena River sites plus sites 55 and 56 from this study (Fig. 1). Close to Douglas Channel,

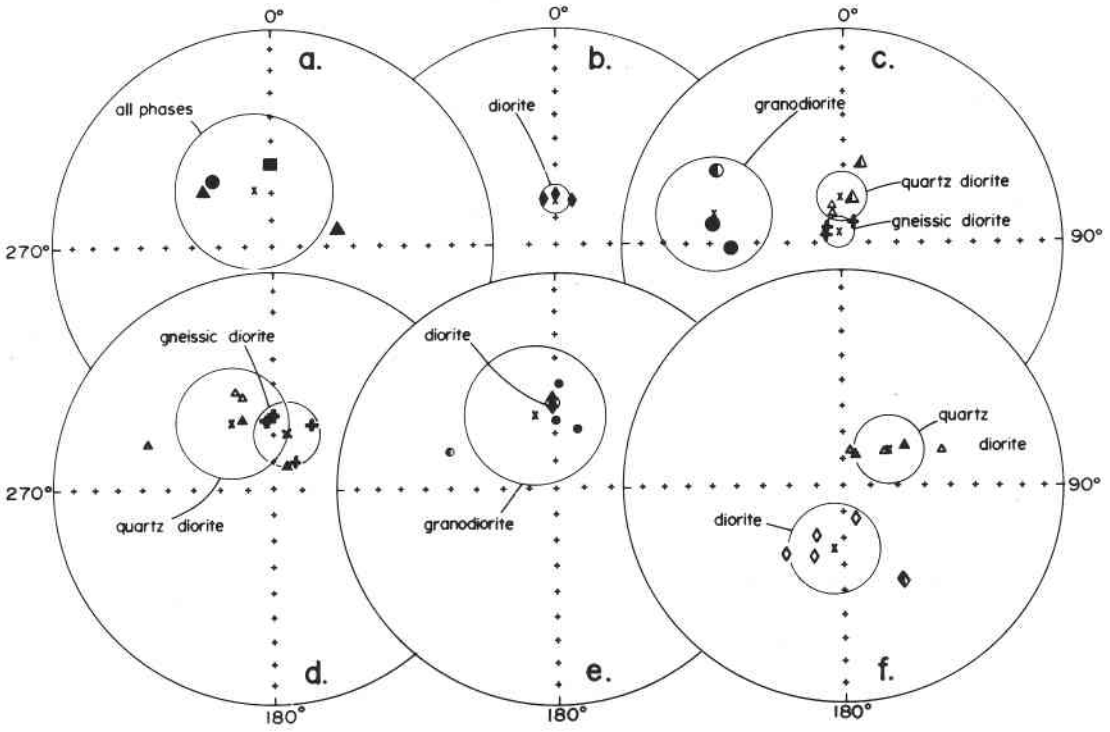


FIG. 4. Equal-area projections showing site mean remanence directions for the (*a*) Ponder, (*b*) Kasiks, (*c*) Quottoon, (*d*) and (*e*) Butedale, and (*f*) Ecstall plutons. Symbols as in Fig. 2 with half-filled symbols indicating antiparallel directions and X the phase mean direction surrounded by its cone of 95% confidence.

the trend swings at a hinge line to the WNW (N 75° W) trend of the centre limb through Hawkesbury Island. The centre limb swings at its eastern hinge to the dominant NNW trend of the Coast plutonic complex to form the south limb.

Site 16 has an anomalous site mean direction, with an intermediate negative inclination to the northwest. This direction is intermediate between those of sites 14 and 15 with poorly defined normal directions, and sites 17 and 18 with well defined reversed directions. Site 16 may record the midpoint of a polarity transition in the earth's magnetic field. It could also be significant that site 16 is close to a phase contact. Regardless, site 16 is anomalous and its direction has been excluded from further statistical analysis.

When the site mean directions are combined (Table 1), they are consistent within one phase on one limb of the Hawkesbury Warp. For example, the Kasiks diorite (Fig. 4), the Quottoon gneissic diorite (Fig. 4*c*), and the Ecstall diorite and quartz (Fig. 4*f*) on the north limb give consistent directions including both Kitimat and Skeena River sections. The site mean directions in one phase from the north limb are consistent with those from the south limb, *e.g.* the Quottoon quartz diorite phase (Fig. 4*c*). When site mean directions from the centre limb are included in the average, the scatter of the directions increases significantly and their mean swings from northeasterly to northwesterly, *e.g.* the Ponder phases (Fig. 4*a*), the Quottoon granodiorite (Fig. 4*c*), the Butedale gneissic diorite, quartz diorite, and granodiorite (Fig. 4*d,e*). Finally, the phase mean directions from north and south limbs are consistent and statistically indistinguishable for one pluton, whereas the direction from the centre limb is statistically different and less well defined. This is true for both the Quottoon and Ecstall–Butedale plutons.

The site mean directions were next grouped by pluton on structural grounds and the populations of the NNW and WNW trends compared using the angular variance ratio test (Larochelle 1969) (Tables 2 and 3; Fig. 5). The first

TABLE 2. Mean remanence directions for groups of sites

Group	Sites*	Limbs†	No. of sites	Mean remanence direction‡ Length R	Decl.	Incl. +down	α_{95}
By petrologic phase							
(1) Kasiks diorite	*5–7*	N	3	2.993	359	73	5
(2) Quottoon gneissic diorite	*13, 15, 16*	N	3	2.989	343	84	6
(3) Quottoon quartz diorite	17, 18, *9–11*	N, S	5	4.948	1	72	9
(4) Quottoon grano-diorite	9–12	C	4	3.870	291	38	20
(5) Butedale gneissic diorite	29, 38, 47, 78	C, S	4	3.937	10	68	13
(6) Butedale diorite	20, 37	C, S	2	1.999	356	56	4
(7) Butedale quartz diorite	13, 31–33, 78	C, S	5	4.714	328	60	21
(8) Butedale grano-diorite	23, 26, 74, 75	C, S	4	3.758	344	62	27
(9) Ecstall diorite	53, *17–20*	N	5	4.788	9	66	18
(10) Ecstall quartz diorite	55, 56, *22–24*	N	5	4.880	50	69	13
By structural limb							
(11) Ponder–Kasiks§ north	*5–7*		3	2.993	359	73	5
(12) Ponder–Kasiks centre	1, 4, 6		3	2.739	333	70	48
(13) Quottoon north + south	17, 18, *9–11 13, 15, 16*		8	7.898	357	77	7
(14) Quottoon centre	9–12		4	3.870	291	38	20
(15) Eastern Zone north + south	17, 18, *5–7 9–11, 13, 15, 16*		11	10.884	358	76	5
(16) Eastern Zone centre	1, 4, 6, 9–12		7	6.251	301	52	23
(17) Ecstall–Butedale north + south	20, 23, 47, 55, 56, 74, 75, 78, *17–20, 22–24*		15	14.584	26	67	7
(18) Ecstall–Butedale centre	13, 26, 28–33, 37, 38, 42, 46, 53, 54		14	13.500	336	59	8

*Italic site numbers refer to the Skeena River Section (Symons 1974*a*).
†Limbs of the Hawkesbury Warp: N, north; C, centre; and S, south.
‡All reversely polarized sites have been changed to their antiparallel position to calculate the mean.
§Includes Alastair Lake pluton sites.

test compares the north limb to the centre limb for the Ponder – Alastair Lake – Kasiks plutons. The test is positive indicating that both limbs have similar populations of directions because the variance ratio of 0.26 is less than the theoretical statistics of 4.46. The test result is not very reliable because both populations are very small. Also the direction for site 1 is divergent for the centre limb, perhaps because site 1 is located in a slightly sheared outcrop which might indicate tectonic effects, or because the WNW trend is not readily apparent at site 1 and possibly it should be included in the NNW limb. If site 1 is excluded, the variance becomes 41.3 so that the test becomes negative indicating that the two populations are not the same. For the Quottoon pluton the test involves 12 sites and is significantly negative. The variance ratio of 37.7 is much greater than the theoretical statistic of 3.5 indicating that the directions from the NNW limb form a different population from those from the WNW centre limb. When all site mean directions from the eastern K–Ar age zone are combined, including the errant site 1 direction, the variance ratio of 11.2 greatly exceeds the theoretical statistic of 3.3. Finally

TABLE 3. Angular variance ratio analysis for the Hawkesbury Warp regional fold test

Test units compared	Number of sites—B	Variance ratio δ_b^2/δ_w^2	Theoretical statistic $F_{0.05}$	Correlation test result
Ponder–Alastair–Kasiks:	6	0.26	4.46	Positive
north *vs.* centre limb	5*	41.50	5.14	Negative
Quottoon: north + south *vs.* centre limb	12	37.73	3.49	Negative
Eastern Zone: north + south *vs.* centre limb	18	11–16	3.30	Negative
Ecstall–Butedale: north + south *vs.* centre limb	29	17.05	3.17	Negative

NOTES: The variance ratio δ_b^2/δ_w^2 is the between-unit over the within-unit angular variances in (radians)^2. The theoretical statistic is $F_{0.05} = F_{2,2(B-1),0.05}$, thereby setting the test at the 95% confidence level. The asterisked result is discussed in the text.

the Ecstall–Butedale pluton of the central K–Ar age zone gives the same highly significant negative test result involving 29 site mean directions. The variance ratio of 17.1 for the NNW versus WNW trends greatly exceeds the theoretical statistic of 3.2. It is clear therefore that the remanence isolated in the plutons predates the formation of the Hawkesbury Warp.

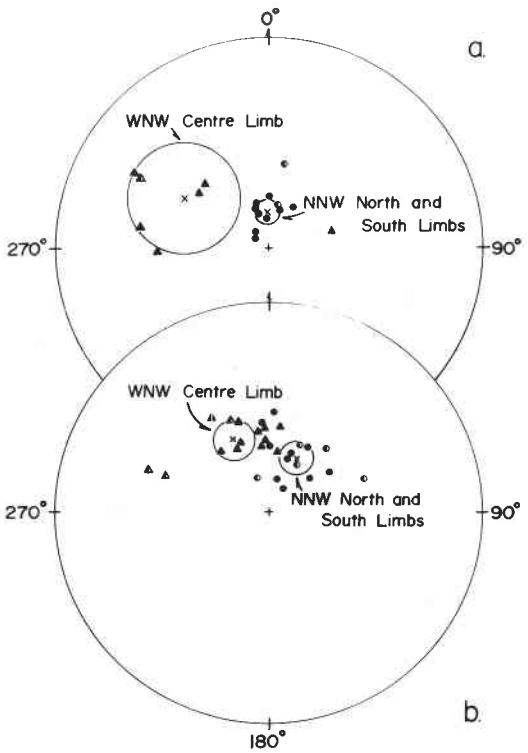


FIG. 5. Site mean remanence directions for the WNW centre limb (triangle) and NNW north and south limbs (circles) of the (*a*) eastern age zone plutons, and (*b*) central age zone plutons. Symbols as given in Figs. 2 and 4.

Pole Positions

To unfold poles around the limbs of the Hawkesbury Warp, it is necessary to compare them to reference pole positions for contemporaneous rock units in tectonically stable portions of the North American craton. The pole positions are listed in Table 4 and illustrated in Fig. 6.

Eastern Age Zone

The reference pole for the 40–50 Ma eastern age zone is the average of 7 poles, *i.e.* the 30 ±5 Ma Oligocene San Juan Mountain igneous suite of Colorado (Diehl *et al.* 1974) at 62° W, 86° N, the 30 ±8 Ma Oligocene volcanics of west-central U.S.A. (Grommé and McKee 1971) at 214° W, 78° N, the 40 Ma Spanish Peak dikes of Colorado (Larson and Strangway 1969) at 149° W, 81° N, the Eocene Absaroka volcanics of Wyoming (Pruss and Shive 1975) at 205° W, 87° N, the 47 ±2 Ma felsite intrusions of Virginia (Løvlie and Opdyke 1974) at 315° W, 87° N (see also Fullager and Bottino 1969), the 50 Ma Green River red beds (Torreson *et al.* 1949) at 168° W, 85° N, and the 58 ±2 Ma Cape Dyer lavas (Deutsch *et al.* 1971) at 55° W, 83° N. There are several poles from the North American Cordillera in the same 30–60 Ma period which give concordant positions including the Mary's Peak sill (Clark 1969), the Mt. Washington stock (Symons 1971*a*), the Hope granitic pluton (Symons 1973*a*), and the East Sooke gabbro (Symons 1973*b*).

The pole position for the NNW trend in the eastern age zone of the Cordillera is coincident

TABLE 4. Summary of pole positions

Group	Pole position Long. °W	Pole position Lat. °N	95% confidence limits Semi-axis δ_p°	95% confidence limits Semi-axis δ_m°	95% confidence limits Radius α_{95}°
Eocene: North American craton*	164	87	—	—	5
Eastern age zone—NNW trend	134	81	8	9	—
—WNW trend	221	44	22	31	—
Late Cretaceous: North American craton*	176	77	—	—	12
Central age zone—NNW trend	37	74	9	11	—
—WNW trend	250	69	9	12	—

*Basis of calculation given in the text.

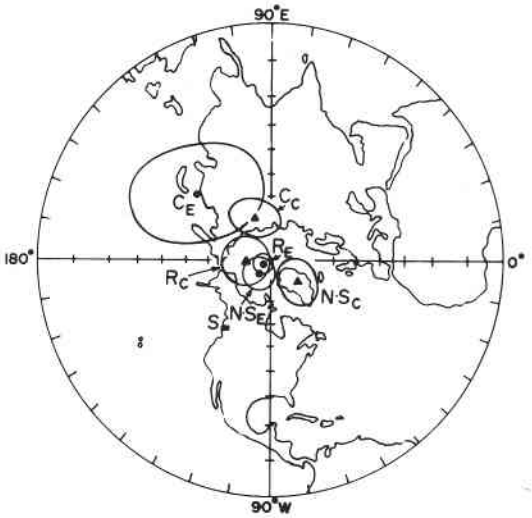


FIG. 6. Pole positions as listed in Table 4 with their ovals of 95% confidence. The reference pole, north plus south limb pole, and centre limb pole for the eastern and central age zones are indicated by R_E, R_C, N–S_E, N–S_C, C_E, and C_C respectively. S is the sampling area.

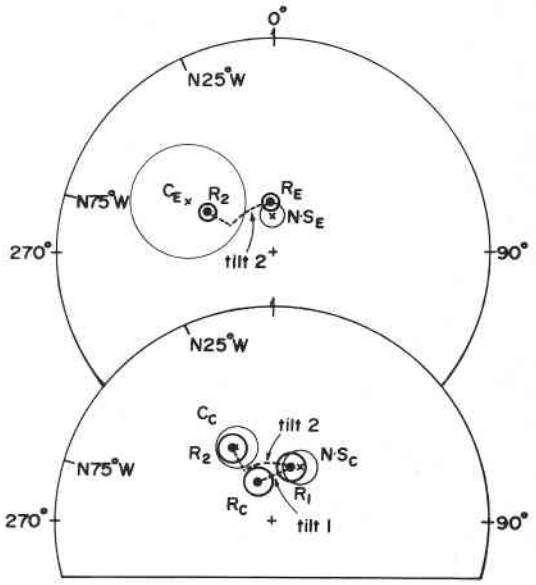


FIG. 7. Movement of reference remanence directions on deformation. Symbols as in Fig. 6. R_1 is the position of the reference direction after a ≈15° westerly tilt (tilt 1), and R_2 is the position expected for the centre limb of the Hawkesbury Warp after ≈50° counterclockwise rotation and ≈10° upward tilt (tilt 2).

with its reference pole for the North American craton. Thus Hawkesbury Warp deformation has rotated only the centre limb pole. The remanence directions indicate that the centre limb has been rotated ≈50° counterclockwise and tilted up at the west end by ≈10° (Fig. 7). The counterclockwise rotation is consistent with the field evidence because the eastern age zone plutonic trend undergoes a similar ≈50° rotation. The limited evidence for the relative ≈10° upward tilt of the northwest end of the centre limb in the eastern age zone plutons will be discussed along with the evidence for such a tilt in the central age zone. The most likely time of deformation would be during the terminal stages of the orogenic activity related to the emplacement of the plutons about 40 Ma ago. Following the Atwater (1970) model for the tectonic evolution of Cenozoic plates on the western margin of North America, a second possible time would be about 27 Ma ago. At this time a triple junction of three ridges separating the Pacific, Kula, and Farallon plates was presumably located at the subducting western edge of the overriding North American plate at about the 53° N latitude of the Hawkesbury Warp. Jostling of the overlying tectonic blocks in the North American plate could be caused by differential motions around the triple junction in the underlying subducting plates.

The Miocene and Recent igneous events in the Prince Rupert – Kitimat area appear to be too minor to be related to such a major deformation and presumably postdate it.

Central Age Zone

The reference pole for the 64–80 Ma central age zone is the average of 6 poles, *i.e.* the 58 ±2 Ma Cape Dyer lavas (Deutsch *et al.* 1971) at 55° W, 83° N, the 63 ±3 Ma volcanics and sills in the Colorado Front Ranges (Hoblitt and Larson 1975) at 273° W, 80° N, the 75 ±7 Ma Boulder batholith in Montana (Hanna 1967) at 193° W, 76° N, the 79 ±10 Ma Niobara limestone of west-central U.S.A. (Shive and Frerichs 1974) at 167° W, 62° N, the 80 Ma Elkhorn Mountain volcanics of Montana (Hanna 1967) at 171° W, 69° W, and the 97 ±10 Ma Arkansas alkalic complexes (Scharon and Hsu 1969) at 170° W, 70° N. The poles for the Sierra Nevada pluton (Grommé and Merrill 1965) and the Howe Sound plutons (Symons 1973*c*) in the Cordillera are contemporaneous and concordant with the cratonic reference pole.

The pole position for the NNW trend in the central age zone of the Cordillera is not coincident with the reference pole for the craton. Therefore it must have been rotated before intrusion of the eastern age zone plutons. After diapiric intrusion of the Ecstall–Butedale pluton (Hutchison 1967) (Figs. 7, 8) some 69 ±5 Ma ago it was tilted to the southwest by ≈15° about a horizontal NNW–SSE axis (Figs. 7, 8*b*). Thus the primary late Upper Cretaceous magnetization was rotated to the present NNW trend direction. Ample evidence for such a tilt comes from faults, foliations, and contacts on the perimeter of the pluton which consistently strike NNW–SSE and dip between 60° NE and 90° (Roddick 1970; Hutchison 1967) with a 15° W tilt reflecting an average deviation from the vertical. Two tectonic hypotheses may explain this tilt in the 69–45 Ma time interval. First, following a variant of the Atwater (1970) model, the westerly moving North American continental plate could have obliquely overriden the northerly moving Kula oceanic plate to form an easterly-dipping subduction zone. The tectonic panels on the western margin of the North American plate would then be tilted to the west. Geochemical evidence favouring an easterly-dipping subduction zone under the Prince Rupert and Kitimat sections has been given by Hutchison (W. W. Hutchison, personal communication, 1975) and by Smith (T. E. Smith, personal communication, 1975). This hypothesis fits into the plate tectonic model proposed by Monger *et al.* (1972) for the Canadian Cordillera. The second hypothesis is that the Ecstall–Butedale pluton was pushed over in response to diapiric intrusion of the adjacent and younger Quottoon pluton along its easterly side. Both hypotheses could be valid if the mechanisms were active either in concert or independently.

After emplacement of the Quottoon pluton, the Hawkesbury Warp deformation rotated the centre limb remanence directions in the Ecstall–Butedale pluton by ≈50° counterclockwise and tilted up its west end by ≈10°. Considering the error limits, these angular values are statistically the same as those determined for the eastern age zone both in sense and magnitude. The ≈50° counterclockwise rotation (Fig. 8*c*,*d*,*e*) is consistent with the ≈50° rotation in the outcrop pattern between the NNW and WNW trends. The outcrop distribution provides good evidence for the upward tilt of the west end of the centre limb (Fig. 11*e*). The Ecstall pluton terminates with a tadpole-shaped head near Prince Rupert exposing a high level monzonite–granodiorite core. Going southerly over 100 km to the hinge line at Douglas Channel, the pluton tapers to its tail exposing increasingly intermediate level quartz diorites thereby indicating that the tectonic panel has been elevated 'up' at the hinge line. Changing strike, the pluton bloats out rapidly over the 25 km of the centre limb to form the northern end of the Butedale pluton and high level granodiorites become the dominant rock type. Clearly the tectonic panel is relatively 'down' at the eastern hinge line. Continuing 70 km southward the Butedale pluton returns to NNW trend, starts to taper down in width, and increasingly exposes intermediate level quartz diorites so the tectonic panel has been progressively raised 'up'. Because the north and south limbs are many tens of kilometres long, the rotations of their remanence directions by the 'up' or 'down' tilt is not statistically determinate. Conversely, the centre limb is short so that a few km (≈4) difference in exposure level over a 25-km limb length would give the required ≈10° tilt. Finally, the Quottoon pluton of the eastern age zone has intermediate level quartz diorite exposed throughout most of its 200 km length. This could reflect a more basic

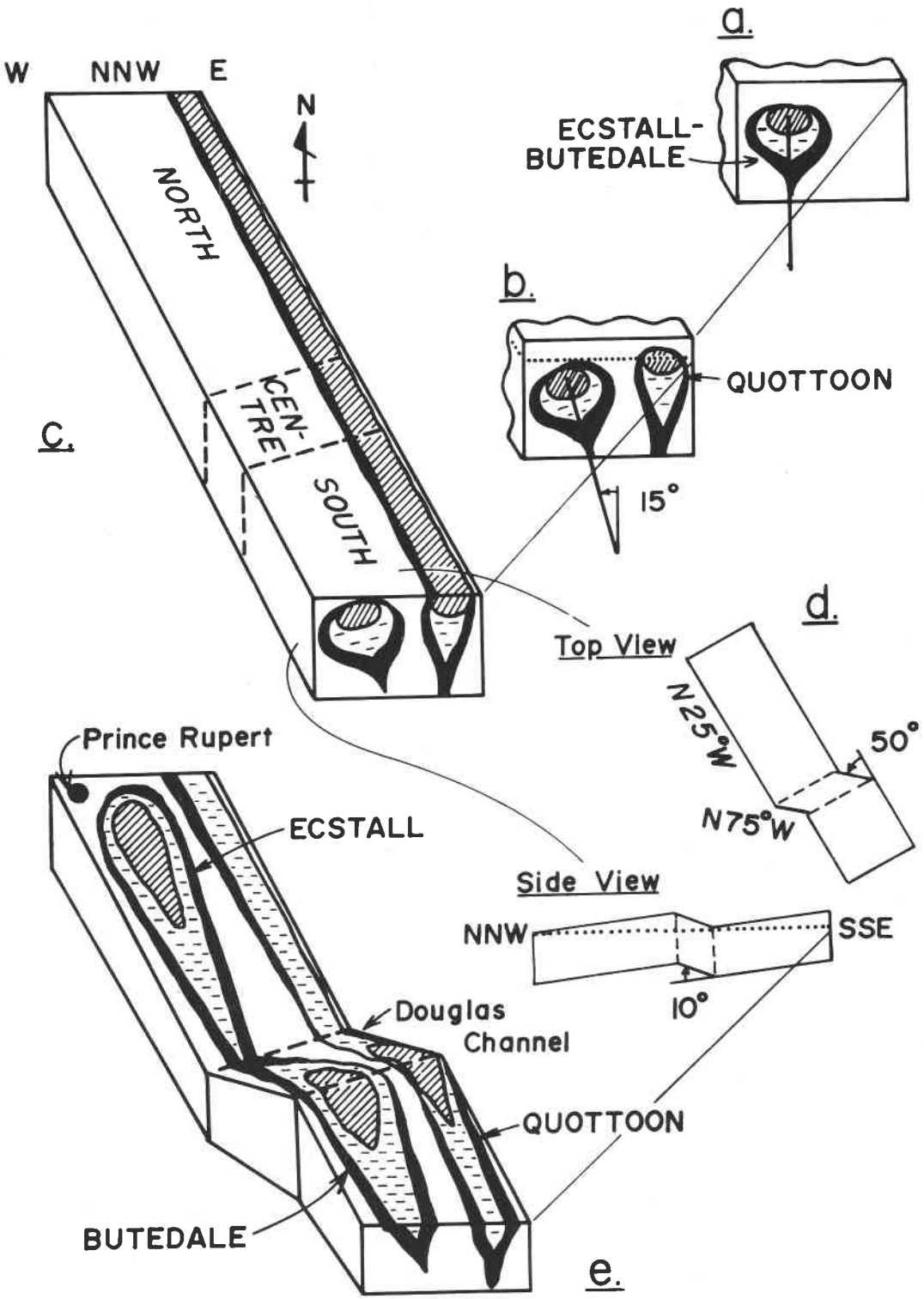


FIG. 8. Schematic of the tectonic events as described in text.

magma, emplacement at a higher level, or greater uplift of the Quottoon pluton relative to the Ecstall–Butedale pluton. Nevertheless, it is consistent with the proposed tectonic model that the only major exposure of higher level granodiorites along the pluton is located around the hinge line between the centre and south limbs.

Pattern of Reversals

Cox (1968, 1969), Kono (1973), and Irving and Pullaiah (1976) have shown that the average interval between polarity changes in the earth's magnetic field during the last 80 Ma has been in the order of ≈ 0.25 Ma, that the maximum duration is ≈ 1 Ma, and that the total times of normal and reverse polarities are about equal. In the Skeena River section (Symons 1974*a*), each plutonic phase was found to exhibit only one polarity except for the Ecstall diorite phase. Either progressive regional uplift from west to east or passage of a thermal metamorphic event from west to east would cause the isotherms to migrate at a reasonably uniform rate from west to east. While either mechanism could yield the K–Ar age zone distribution, it would necessarily yield a crudely alternating polarity pattern to the site remanence directions along an east–west section. Because the pattern was not found, it was argued that the K–Ar dates and the paleomagnetic directions record the time of emplacement cooling of the plutons. Further they must have cooled rapidly within one or at most two polarity intervals or within ≈ 0.5 Ma.

This study provides further enlightenment. The Hawkesbury Warp and the earlier tilting of the Ecstall–Butedale pluton are the major evident tectonic events and they should be related to any hypothetical major thermal metamorphic or regional uplift event. However the magnetization clearly predates these tectonic events which makes the thermal metamorphic or regional uplift hypotheses for forming the remanence untenable. The pattern of a single polarity in each pluton or plutonic phase is fairly well maintained (Fig. 9). All of the Ponder and Alastair Lake pluton sites are normal. The Quottoon pluton gives polarity transition at its south end (sites 14 to 18, Figs. 1, 9, Table 1), and at site 11 on Douglas Channel. Its 7 sites on the Skeena River section are all normal. Similarly the Ecstall–Butedale pluton has three reversed sites (13, 32, 46) in the centre limb—all other sites are normal except near Prince Rupert where its remanence is dominantly of reversed polarity with two adjacent normal sites. Summarizing, the plutons do not show an alternating remanence polarity pattern in sites along the section thereby further supporting the concept that they acquired their remanence during emplacement rather than subsequently during some metamorphic event or during progressive uplift. Finally, it appears that the plutons were intruded and cooled through their blocking temperature range of 500–600°C over 2–4 polarity intervals of ≤ 1 Ma total duration rather than the 1–2 intervals as previously proposed.

Geotectonic Significance

Several hypotheses have been advanced to explain discordant paleomagnetic pole positions determined for rock units in the Cordillera, including: (*a*) a rapid excursion of the geomagnetic pole from its usual position during the time of remanence acquisition (*e.g.* Grommé and Gluskoter 1965; Saad 1969); (*b*) severe local deformation during orogenesis (*e.g.* Symons 1974*b*; Whitney 1975); (*c*) major regional tectonic rotations extending over several tens of kilometres (*e.g.* Watkins 1965; Symons 1971*d*, 1973*d*); (*d*) plate tectonics causing large scale—>3000 km—northward translation of the western Cordillera relative to the North American craton (*e.g.* Irving and Yole 1972; Beck and Noson 1972; Packer and Stone 1972); and, (*e*) plate tectonics causing substantial northward translations on right-lateral shears and clockwise-rotations in the western-most Cordillera (*e.g.* Beck 1976).

A major difficulty with the plate tectonic hypotheses—(*d*) and (*e*)—is explaining the many concordant pole positions from the Cordillera. These include poles from the westernmost Cordillera in the Insular Belt dating back to Jurassic (Symons 1971*a*,*b*, 1973*b*) and possibly Triassic time (Symons 1971*c*), in the Cascade and Coast Ranges back to Jurassic time (Beske *et al.* 1973; Symons 1969*a*, 1973*a*, 1973*c*), and in the Interior Plateau back to Jurassic time (Symons 1969*b*, 1973*e*). A further difficulty with (*e*) is the existence of poles indicating counterclockwise rotations (*e.g.* Grommé *et al.* 1967).

It is unreasonable to invoke excursions of the geomagnetic field—*i.e.* (*a*) above—to explain discordant poles in opposite directions from the reference pole from one pluton as found in the

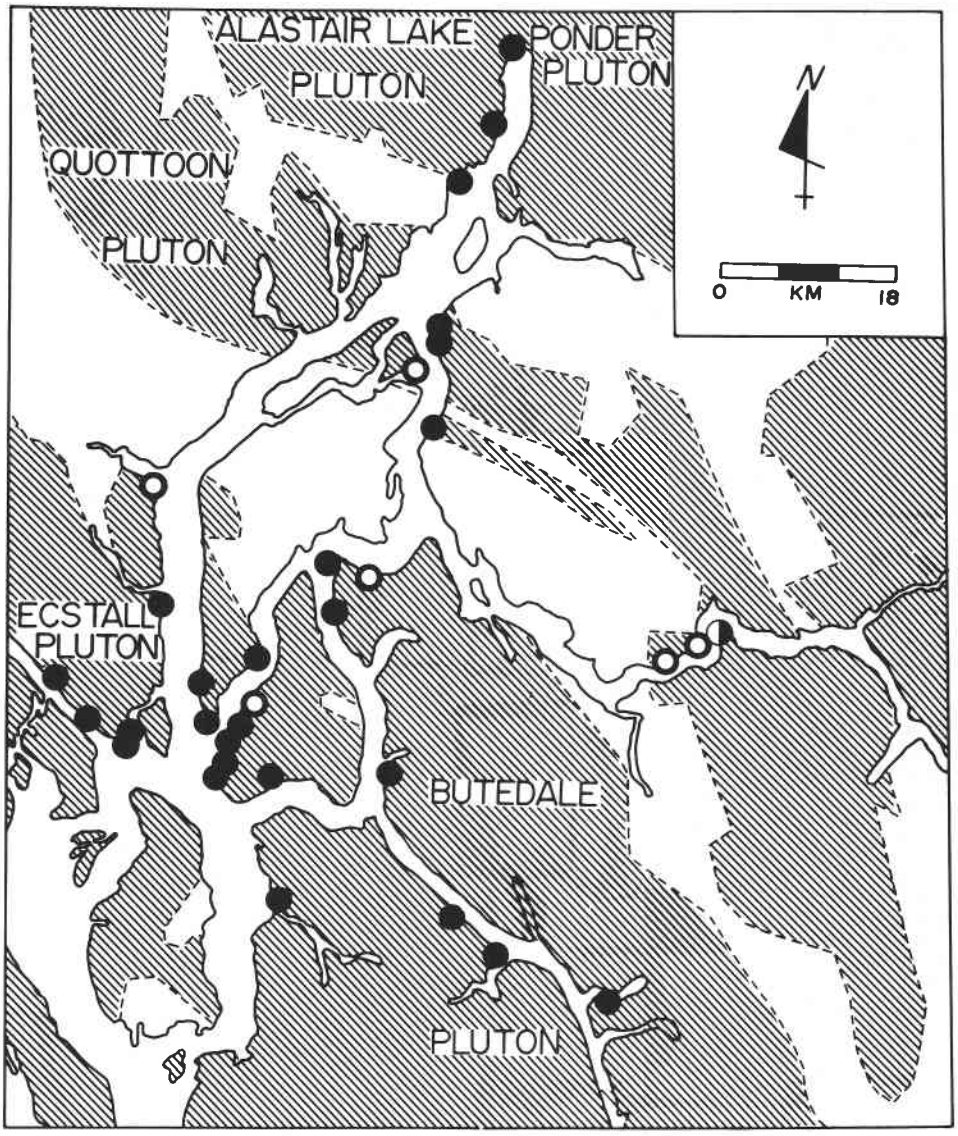


FIG. 9. Remanence polarity of sites indicating normal sites as solid circles, reversed sites as open circles, and the transitional site as a half-open circle.

Butedale pluton, or to explain discordant and concordant poles from one pluton as found in the Quottoon pluton. The size of the area involved in this study excludes alternative (*b*) above. It is obviously inconsistent to explain the different remanence inclinations found in the various limbs by different amounts of northward translation so that alternative (*d*) is invalid. Finally, the concept of northward translation on shears combined with clockwise rotation—alternative (*e*)—explains neither inclinations nor declinations because the centre limb poles reflect varying amounts of counterclockwise rotations whereas the adjacent north and south limbs reflect either no rotation or minor amounts of clockwise rotation. The paleomagnetic results from the Prince Rupert – Kitimat area plutons can only be explained in terms of regional tectonic rotation—*i.e.* (*c*) above as previously described.

Conclusions

(*1*) Slightly more than one half (32) of 59

sites from the Kitimat section have a stable well grouped within-site remanence direction after af demagnetization. These include 100% of the hornblende diorite sites, 63% of quartz diorite sites, 50% of the granodiorite sites, 45% of the gneissic diorites, and 20% of the monzonite sites. Stable sites were found for all plutons sampled–Ponder, Alastair Lake, Quottoon, Ecstall, and Butedale.

(*2*) The site mean directions are statistically consistent within one phase on one limb of the Hawkesbury Warp, within one phase between north and south limbs, and within one pluton between north and south limbs.

(*3*) The site mean directions predate the deformation of the S-shaped Hawkesbury Warp and provide a regional fold test in sites spread along a 200-km strike length.

(*4*) The pole for the NNW trending north and south limbs in the 40–50 Ma eastern K–Ar age zone is concordant with a reference pole calculated from contemporaneous rock units located in the North American craton. The pole for the WNW trending centre limb is discordant reflecting a $\approx 50°$ counterclockwise rotation and a $\approx 10°$ relative upward tilt of the west end of the limb. The strike trends and distribution of plutonic phases agree with these rotations.

(*5*) Hawkesbury Warp deformation occurred after emplacement of the eastern age zone plutons some 40 Ma ago either as a terminal orogenic event or in response to plate motions about the Pacific–Kula–Farallon triple junction at the North American plate boundary some 27 Ma ago.

(*6*) The pole for the NNW trending limbs of the 69 ± 5 Ma Ecstall–Butedale pluton in the central K–Ar age zone is discordant with the North American cratonic reference pole. The pole reflects a post-intrusion $\approx 15°$ tilt of the pluton towards the southwest about a horizontal NNW–SSE axis. Abundant evidence for the tilt comes from steep eastward dips of foliation, contact, and fault planes along the flanks of the pluton. The tilt occurred before emplacement of the adjacent Quottoon pluton, *i.e.* between 69 ± 5 Ma and 45 ± 5 Ma. The tilt was probably related either to deformation in the westerly-moving continental crust of the North American plate as it obliquely overrode the northerly-moving oceanic crust of the Kula plate to form an easterly-dipping subduction zone, or to diapiric intrusion of the Quottoon pluton to the east. Subsequent deformation formed the Hawkesbury Warp with its WNW centre limb rotated $\approx 50°$ counterclockwise and tilted up by $\approx 10°$ at its west end. As before the plutonic strike trends and phase distributions strongly support this tectonic interpretation.

(*7*) The Kitimat section does not show the alternating polarity pattern which would be produced if the plutons acquired their remanence from easterly migrating isotherms caused by either progressive regional uplift from west to east or passage of a regional thermal metamorphic event. Both mechanisms could explain the distribution of K–Ar age dates in the central and eastern age zone, but they could not produce the observed polarity pattern. Both the K–Ar ages and the site mean directions date the initial cooling of the plutons. The positive fold test indicates that the remanence is predeformational providing additional evidence against the uplift and the thermal metamorphism hypotheses.

(*8*) Individual plutons record a maximum of 1–4 geomagnetic polarity intervals over their sampled length of up to 170 km indicating that they were emplaced and cooled down to $\approx 400°C$ within about 1 Ma.

(*9*) The discordant pole positions cannot be explained by invoking an excursion of the geomagnetic paleofield, a large scale >3000 km northward translation of the western Cordillera relative to the North American craton, or a combination of substantial clockwise rotations and northward translations on major shear faults along margin of the North American plate. The poles can be uniquely explained in terms of regional geotectonics.

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